

# From live conversation to verified action

A focused benchmark of Waymark's real-time speech, reasoning, routing and action pipeline.



## Benchmark question

Can Waymark turn Nigerian English and Pidgin call transcripts into grounded, policy-controlled work while the conversation is still happening?

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### CORE TESTS PASSED

API, calling, actions, routing, security and learning checks.

**3 sec**

### LIVE AUDIO WINDOW

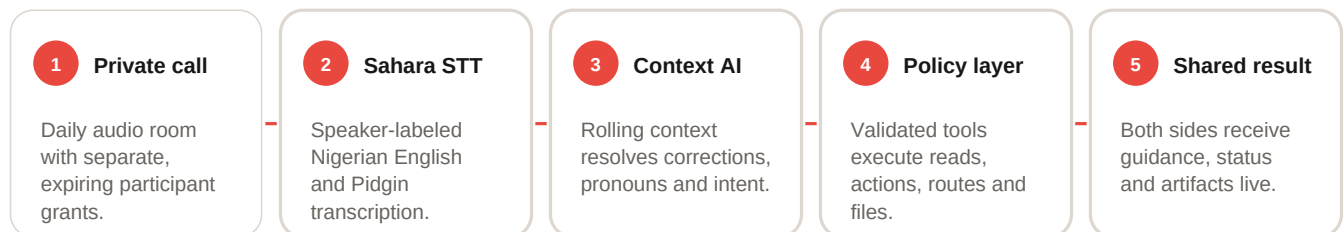
Speech is committed incrementally so analysis starts before hang-up.

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### SANDBOX VERTICALS

Delivery, banking, telecom and business workflows are exercised.

## SYSTEM UNDER TEST



## HEADLINE FINDING

The current build validates the full execution path: connect two speakers, capture role-labeled audio, interpret route or service intent, enforce confirmation rules, persist an audit trail, and publish the result during the call. The strongest evidence is functional reliability. Comparative speech accuracy remains an early smoke test.

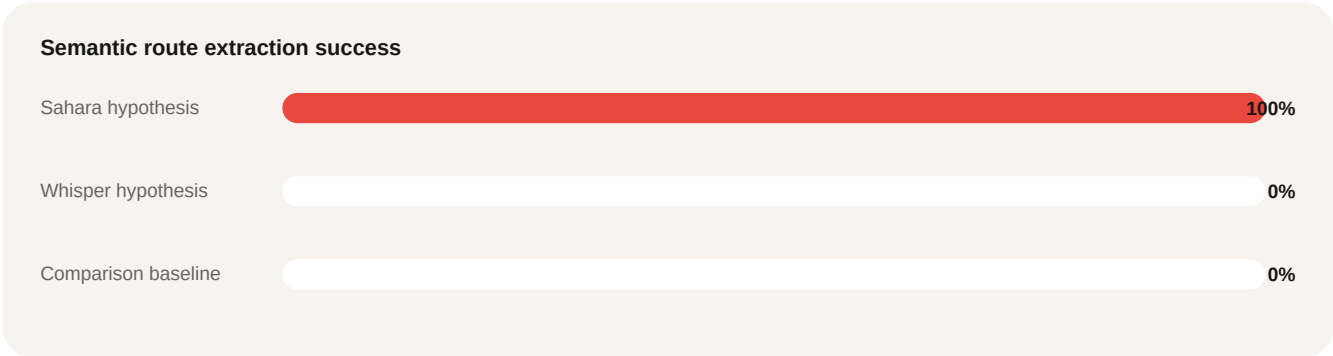
# Speech-to-route smoke benchmark

Two labeled Nigerian delivery scenarios test whether transcript differences preserve the route meaning.

METHOD

The repository corpus contains one standard English route and one Pidgin-English code-switched route. Each transcript hypothesis is scored against the reference for word error rate (WER), landmark recall, relation accuracy, complete route extraction and recorded latency. The same deterministic evaluator is used for every row.

| Hypothesis          | WER (lower) | Landmark recall | Relation accuracy | Complete route | Mean latency |
|---------------------|-------------|-----------------|-------------------|----------------|--------------|
| Sahara              | 0.0%        | 100.0%          | 100.0%            | 100.0%         | 605 ms       |
| Whisper-style       | 27.0%       | 33.3%           | 50.0%             | 0.0%           | 895 ms       |
| Comparison baseline | 35.1%       | 33.3%           | 100.0%            | 0.0%           | 745 ms       |



WHAT THE SAMPLE SHOWS

The Sahara reference hypotheses preserve every labeled landmark and spatial relation across both examples, enabling complete route extraction. Competing hypotheses preserve some intent but lose named or descriptive landmarks.

WHAT IT DOES NOT SHOW

These are embedded transcript hypotheses, not fresh provider inference over recorded audio. With n=2, the table validates metric plumbing and semantic sensitivity only. It is not a production accuracy or provider ranking claim.

CODE-SWITCHED EXAMPLE

*"When you reach FirstBank, pass am small, second street by your left, na the blue gate beside the church."*

# Functional evidence and next benchmark

Automated checks cover the execution path around speech, including access, action safety and persistence.

AUTOMATED VALIDATION

19 tests | 3.85 seconds | 100% pass

|                   |                                                                             |      |
|-------------------|-----------------------------------------------------------------------------|------|
| Call access       | Private Daily links, role grants and join cookies are validated.            | PASS |
| Audio pipeline    | Speaker tracks, PCM conversion and provider media formats are checked.      | PASS |
| Conversation AI   | Corrections are applied and unsupported landmarks are rejected.             | PASS |
| Delivery guidance | Address search, map candidates, route steps and learning are exercised.     | PASS |
| Agent actions     | Customer lookup, guarded account actions and invoice creation complete.     | PASS |
| Security          | Tampered access tokens, provider signatures and authorization are rejected. | PASS |

DEPLOYMENT READINESS OBSERVED

LIVE CONFIGURATION

Production health reports HTTPS, PostgreSQL, Daily, Sahara, Mapbox and OpenAI credentials present.

RESPONSIBLE EXECUTION

Model storage is disabled, raw audio is not retained, and sensitive tools require confirmation.

REQUIRED NEXT-STAGE BENCHMARK

- 01

Record 50-100 consented calls across Nigerian English, Pidgin and major accents.
- 02

Report WER, semantic task accuracy and action completion by language mix and noise level.
- 03

Measure p50 and p95 latency from speech commit to visible route or completed action.
- 04

Audit false actions, clarification quality, confirmation compliance and user trust.

CONCLUSION

Waymark has a working, test-backed path from a live two-sided call to a controlled downstream result. The next milestone is statistical validation on a larger, consented Nigerian audio corpus under real network and background-noise conditions.